

Alice Blanche Ekane

Senior AWS Infrastructure Engineer/ Site Reliability Engineer (SRE)

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Senior AWS Infrastructure Engineer with 6+ years of experience architecting, automating, and securing cloud environments across healthcare, enterprise, and managed service platforms. Proven expertise in AWS services including EC2, VPC, S3, IAM, Lambda, CloudWatch, CloudTrail, EKS, WAF & Shield, Organizations, Secrets Manager, KMS, Inspector, Cost Explorer, and Trusted Advisor. Skilled in infrastructure automation with Terraform, CloudFormation, and Ansible, and in CI/CD pipelines using Jenkins, GitHub Actions, GitLab, and Kubernetes. Strong programming background with Python (Boto3 for AWS automation: XML, JSON), Django, Django REST Framework, FastAPI, Pandas, and data visualization. Experienced in building fullstack applications, developing APIs, and applying deep learning, large language models, and OpenAI SDK for intelligent monitoring and anomaly detection. Recognized for reducing cloud costs, improving uptime, and enhancing operational efficiency through scalable architectures, proactive monitoring, and secure automation workflows.

Skills

Cloud Platforms: AWS (EC2, VPC, S3, IAM, CloudWatch, CloudTrail, Lambda, SQS/SNS, Config, GuardDuty, Security Hub, CloudFront, ALB/NLB, RDS/Aurora, Systems Manager, AWS Backup, EKS, AWS Organizations, WAF & Shield, Cost Explorer & Trusted Advisor, Secrets Manager, KMS, Inspector)

Infrastructure & Systems: Infrastructure Provisioning, Server Administration, Linux, Windows, Active Directory, Group Policies, Endpoint Management, Patching, Backup & Recovery

IaC & Automation: Terraform, CloudFormation, Ansible, Bash, PowerShell

DevOps & CI/CD: Jenkins, GitHub, GitLab, Git, CI/CD Pipelines, Docker, Kubernetes, DevOps Practices

Programming & Frameworks: Python (Boto3 for AWS automation: XML, JSON), Django, Django REST Framework, FastAPI, Fullstack Frameworks, Pandas, Data Visualization

AI/ML & Emerging Tech: Deep Learning, Large Language Models (LLM), OpenAI SDK

Security & Compliance: SIEM Monitoring, Vulnerability Management, Threat Detection, Log Analysis, Access Management, Identity Hardening

Networking: TCP/IP, DNS, DHCP, NAT, Subnets, Routing, VPN, Load Balancing, Firewall Configurations, Elastic IP Management

Monitoring & Optimization: CloudWatch Metrics/Logs, CloudTrail, AWS Config, Log Insights, Resource Optimization, Cost Management, Cost Optimization

IT Operations: Incident Response, Troubleshooting, Root-Cause Analysis (RCA), Change & Incident Management, ServiceNow, Jira, SLAs/OLAs

Professional Experience

Senior AWS Infrastructure Engineer/Site Reliability Engineer (SRE) , CVS Health

Mar 2024 - Present | TX

- Architected production AWS environments using EC2 and VPC while integrating CloudWatch Metrics/Logs, improving visibility and reducing incident response time by 6 hours per week saved.
- Engineered secure object storage workflows with S3 and implemented AWS Backup, reducing recovery time by 32% and lowering annual storage costs by \$120K across 3 PB of data.
- Automated infrastructure provisioning with Terraform and implemented GitHub workflows, reducing manual deployment errors by 40% and accelerating provisioning by 32% across 50+ environments.
- Orchestrated identity and access governance using IAM and strengthened Security Hub monitoring, decreasing unauthorized access alerts by 50% and avoiding potential remediation costs of \$50K per year.
- Configured hybrid network setups with ALB/NLB and VPN, enhancing application uptime to 99.5% and reducing latency by 13%, supporting 12,000+ daily requests.
- Executed operational excellence initiatives through Endpoint Management, Incident Response, and troubleshooting workflows, reducing average resolution time by 32% and maintaining 99.9% system availability.
- Implemented Amazon SNS and SQS to enable asynchronous messaging between distributed services, improving workflow

reliability and reducing processing delays by 35%.

- Configured AWS Elastic IPs for production EC2 and VPN gateways, ensuring consistent endpoint connectivity and reducing DNS propagation delays by 40%.
- Implemented static IP assignments for load balancers, improving secure access reliability for 12,000+ daily healthcare transactions.
- Drove AWS resource optimization initiatives that cut annual cloud spend by 18% by identifying underutilized workloads and implementing right-sizing strategies.
- Streamlined DNS configurations within hybrid AWS environments, reducing propagation delays by 40% and improving secure endpoint accessibility for healthcare applications.
- Automated AWS resource provisioning and monitoring with Python Boto3 (XML, JSON), reducing manual intervention by 42%.
- Built APIs using Django REST Framework and FastAPI, accelerating integration workflows by 38%.
- Developed secure fullstack Django applications for cloud metrics visualization, improving compliance reporting speed by 33%.
- Applied Pandas and visualization libraries to analyze AWS usage, increasing cost forecasting accuracy by 29%.
- Experimented with Deep Learning, LLMs, and OpenAI SDK, enhancing anomaly detection efficiency by 26%.
- Architected production AWS environments with EC2/VPC and CloudWatch, reducing incident response time by 35%.
- Implemented AWS WAF & Shield, decreasing malicious traffic incidents by 41%.
- Leveraged AWS Organizations for governance, cutting policy drift by 36%.
- Used Cost Explorer & Trusted Advisor, achieving an additional 18% cost reduction.
- Automated secrets rotation with AWS Secrets Manager, reducing credential errors by 40%.
- Applied AWS KMS for encryption, improving secure data exchange reliability by 32%.
- Conducted proactive scans with AWS Inspector, reducing remediation cycles by 30%.

AWS Infrastructure Engineer, Dell Technologies

Nov 2021 - Feb 2024 | TX

- Provisioned cloud infrastructure with Terraform and integrated CI/CD pipelines, cutting deployment time by 15 hours per month and increasing release frequency to 20+ deployments.
- Implemented directory services using Active Directory and Group Policies, reducing configuration errors by 30% and improving login success for 5,000+ users, avoiding approximately \$25K per year in support overhead.
- Monitored cloud and database services with AWS Config and Log Insights, detecting 95% of anomalies proactively and reducing incident escalations by 26%.
- Enforced vulnerability mitigation using Vulnerability Management, decreasing critical risk exposure by 40%, preventing potential downtime costs across 100+ workloads.
- Managed enterprise compute resources using Linux and Windows, improving patch coverage by 41% and reducing system downtime by 18% across 2,500+ servers.
- Configured networking components including Subnets, Routing, and load balancing, improving application availability and reducing latency by 12%, improving SLA adherence and saving around \$30K/year in penalties.
- Used SNS with SQS subscriptions to decouple application components, increasing fault tolerance and improving message delivery success rates by 28%.
- Managed Elastic IPs within CI/CD pipeline deployments, strengthening compliance posture and reducing rollback incidents by 25%.
- Provisioned static IPs for NAT gateways and directory services, improving application availability and reducing latency by 12%, saving \$30K/year in SLA penalties.
- Implemented proactive cost management practices across multi-account AWS deployments, lowering operational expenses by 22% while maintaining SLA compliance.
- Enhanced DNS routing and subnet management to improve application availability, reducing latency by 15% and strengthening enterprise connectivity.
- Automated provisioning pipelines with Python Boto3, reducing manual configuration errors by 37%.
- Designed APIs with Django REST Framework and FastAPI, improving interoperability by 34%.
- Built Django fullstack applications for visualization, reducing troubleshooting time by 28%.
- Leveraged Pandas for spend analysis, improving optimization accuracy by 31%.
- Researched Deep Learning and LLM models with OpenAI SDK, reducing incident escalations by 22%.
- Deployed EKS clusters, improving release velocity by 36%.
- Configured AWS WAF & Shield, reducing downtime incidents by 33%.
- Managed AWS Organizations, improving compliance reporting by 27%.
- Used Cost Explorer & Trusted Advisor, lowering expenses by 22%.
- Automated credential storage with Secrets Manager, reducing rollback incidents by 25%.

- Applied AWS KMS, ensuring secure communication with 31% fewer encryption errors.
- Conducted vulnerability scans with AWS Inspector, reducing critical exposure by 40%.

Cloud Operations Engineer, DXC Technology

Aug 2019 - Oct 2021 | VA

- Supported operational workflows through structured Incident Response, resolving 95% of incidents within SLA and maintaining system at 98.7%, saving around 8 hours per week in manual recovery.
- Resolved infrastructure disruptions via Troubleshooting, reducing escalations by 25% and improving end-user satisfaction for 10+ client accounts, avoiding approximately \$20K/year in support costs.
- Investigated recurring technical patterns using Root-Cause Analysis (RCA), mitigating 37% of repeat incidents and increasing system reliability.
- Monitored security posture with SIEM Monitoring and Log Analysis, detecting 85% of abnormal activities proactively and reducing potential breach costs by \$45K/year.
- Facilitated change workflows using Jira, decreasing failed deployments by 18% and improving update compliance for 50+ components, avoiding unnecessary downtime costs.
- Configured secure connectivity layers via VPN, enhancing remote-access reliability by 23%, supporting 500+ users and reducing connectivity-related support tickets by 40%.
- Configured Amazon SNS topics and SQS queues to support alerting and background processing, reducing incident response latency by 22%.
- Automated AWS provisioning with Python Boto3, reducing manual recovery efforts by 39%.
- Developed APIs with Django REST Framework and FastAPI, improving workflow automation by 35%.
- Created Django fullstack applications for visualization, reducing RCA time by 27%.
- Applied Pandas to analyze AWS usage, improving forecasting accuracy by 30%.
- Piloted LLM-based anomaly detection with OpenAI SDK, reducing repeat incidents by 24%.
- Configured EKS clusters, improving system reliability by 33%.
- Implemented AWS WAF & Shield, reducing abnormal traffic incidents by 35%.
- Managed AWS Organizations, simplifying governance and reducing policy drift by 28%.
- Leveraged Cost Explorer & Trusted Advisor, saving clients approximately 19% annually.
- Automated credential management with Secrets Manager, reducing audit issues by 31%.
- Applied AWS KMS, ensuring compliance with 29% fewer encryption gaps.
- Used AWS Inspector, reducing repeat incidents by 25%.

Education

Master of Science in Biochemistry University of Douala, Douala, Cameroon	August 2000 – July 2002
Bachelor of Science in Engineering University of Douala, Douala, Cameroon	August 1995 – June 1999

Certifications:

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| • AWS Certified Cloud Practitioner | December 2023 |
| • AWS Certified Solutions Architect – Associate | February 2024 |
- **Additional Information:**
 - **Languages:** English (fluent), French (fluent)